

Questions

MathonGo

Q1 - 2024 (04 Apr Shift 1)

An elastic spring under tension of 3 N has a length a . Its length is b under tension 2 N. For its length $(3a - 2b)$, the value of tension will be _____ N.

Q2 - 2024 (05 Apr Shift 1)

The density and breaking stress of a wire are $6 \times 10^4 \text{ kg/m}^3$ and $1.2 \times 10^8 \text{ N/m}^2$ respectively. The wire is suspended from a rigid support on a planet where acceleration due to gravity is $\frac{1}{3}$ rd of the value on the surface of earth. The maximum length of the wire with breaking is _____ m (take, $g = 10 \text{ m/s}^2$).

Q3 - 2024 (05 Apr Shift 2)

Match List-I with List-II :

	List-I		List-II
(A)	A force that restores an elastic body of unit area to its original state	(I)	Bulk modulus
(B)	Two equal and opposite forces parallel to opposite faces	(II)	Young's modulus
(C)	Forces perpendicular everywhere to the surface per unit area same everywhere	(III)	Stress
(D)	Two equal and opposite forces perpendicular to opposite faces	(IV)	Shear modulus

Choose the correct answer from the options given below :

(1) (A)-(IV), (B)-(II), (C)-(III), (D)-(I)

(2) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

(3) (A)-(II), (B)-(IV), (C)-(I), (D)-(III)

(4) (A)-(III), (B)-(I), (C)-(II), (D)-(IV)

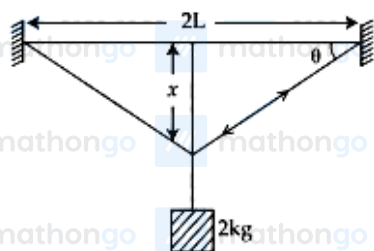
Q4 - 2024 (06 Apr Shift 2)

A wire of cross sectional area A , modulus of elasticity $2 \times 10^{11} \text{ Nm}^{-2}$ and length 2 m is stretched between two vertical rigid supports. When a mass of 2 kg is suspended at the middle it sags lower from its original position

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making angle $\theta = \frac{1}{100}$ radian on the points of support. The value of A is $\underline{\hspace{2cm}} \times 10^{-4} \text{ m}^2$ (consider $x \ll L$).



(given : $g = 10 \text{ m/s}^2$)

Q5 - 2024 (09 Apr Shift 1)

Two persons pull a wire towards themselves. Each person exerts a force of 200 N on the wire. Young's modulus of the material of wire is $1 \times 10^{11} \text{ N m}^{-2}$. Original length of the wire is 2 m and the area of cross section is 2 cm^2 .

The wire will extend in length by $\underline{\hspace{2cm}} \mu\text{m}$.

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Answer Key

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Q1 (5) mathongo Q2 (600) mathongo Q3 (2) mathongo Q4 (1) mathongo ///

Q5 (20) mathongo mathongo mathongo mathongo mathongo mathongo mathongo ///

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Solutions

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Q1

$$3 = K(a - \ell)$$

$$2 = K(b - \ell)$$

$$T = K(3a - 2b - \ell)$$

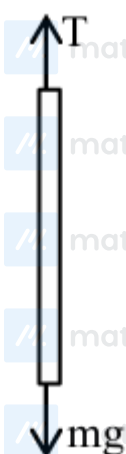
$$T = K(3(a - \ell) - 2(b - \ell))$$

$$= K \left[3 \left(\frac{3}{K} \right) - 2 \left(\frac{2}{K} \right) \right]$$

$$= 9 - 4$$

$$= 5 \text{ N}$$

Q2



$$T = mg$$

$$\sigma = \frac{T}{A} = \frac{mg}{A}$$

$$\Rightarrow \ell = \frac{\sigma}{\rho g} = \frac{1.2 \times 10^8 \times 3}{6 \times 10^4 \times 10} = 600$$

Q3

$$(1) \text{ stress} = \frac{F_{\text{restoring}}}{A}$$

$$\text{If } A = 1$$

$$\text{Stress} = F_{\text{restoring}}$$

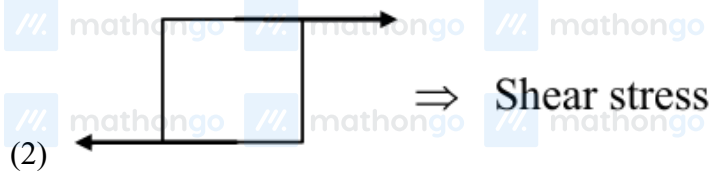
(A)-(III)

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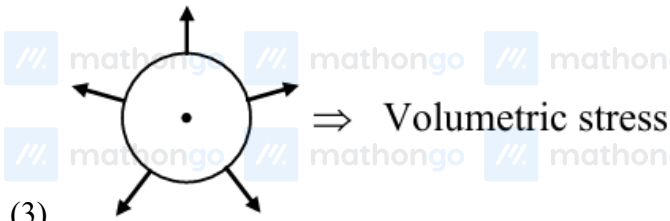
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Solutions

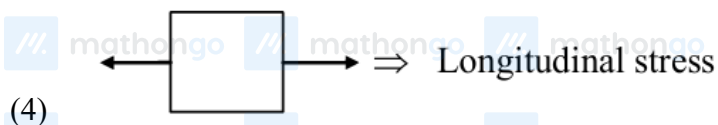
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(B)-(IV)

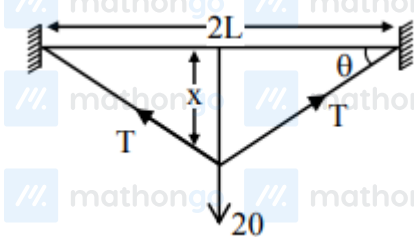


(C)-(I)



(D)-(II)

Q4



In vertical direction

$$2 T \sin \theta = 20$$

using small angle approximation $\sin \theta = \theta$

$$\theta = \frac{1}{100}$$

$$\therefore T = \frac{10}{\theta}$$

$$T = 1000 \text{ N}$$

$$\text{Change in length } \Delta L = 2\sqrt{x^2 + L^2} - 2L$$

$$= 2L \left[1 + \frac{x^2}{2L^2} - 1 \right]$$

$$\Delta L = \frac{x^2}{L}$$

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Solutions

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$$\therefore \text{Modulus of elasticity} = \frac{\text{stress}}{\text{strain}}$$

$$2 \times 10^{11} = \frac{10^3}{A \times \frac{x^2}{L}} \times 2L$$

$$\therefore A = 1 \times 10^{-4} \text{ m}^2$$

Q5

$$200 \text{ N} \longleftrightarrow O$$

$$\frac{F}{A} = Y \frac{\Delta \ell}{\ell} \Rightarrow \Delta \ell = \frac{F \ell}{A Y}$$

$$\Delta \ell = \frac{200 \times 2}{2 \times 10^{-4} \times 10^{11}} = 2 \times 10^{-5} = 20 \mu\text{m}$$

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